

Project Proposal

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1 Introduction

By training for next-word prediction in large datasets of texts, language models learn to encode linguistic properties that can be leveraged in a wide variety of downstream language-related tasks. In particular, transformer-based language models have become a cornerstone technology in Natural Language Processing (NLP) by advancing the state-of-the-art in many NLP tasks. However, their data-hungry nature limits their applicability to the few languages that have sufficiently large datasets for language models to train on.

One solution to this limitation is to have language models train on texts written in a variety of languages. By doing so, such models pick up on near-universal linguistic properties, such as there being a distinction between nouns and verbs [7], as well as properties shared only by subsets of languages, such as verbs preceding their subject in a sentence. This information is then encoded in a shared semantic space that allows multilingual language models to use information obtained from one language when processing another in a process called cross-lingual transfer. This is particularly useful in the low-resource setting: a language model may not have trained on enough data in Scottish Gaelic to have captured that verbs precede their subject in the language. However, if the model has been trained on enough texts in Modern Standard Arabic, it may well have managed to encode that verbs precede their subjects in that language, and consequently transfer that knowledge cross-lingually to better deal with texts in Scottish Gaelic.

In a process called zero-shot transfer, Pires et al. [15] finetune the multilingual language model M-BERT [5] on a source language and evaluate it on a different target language. They report remarkably high performance on the target languages, indicating that multilingual models can be the basis of successful classification models even when no training data is available in a given language. Lauscher et al. [10] report considerable improvements over zero-shot transfer when first finetuning on as few as 10 examples in a target language. Moreover, they find M-BERT's cross-lingual generalizability to be closely linked to how well-represented a target language is in M-BERT's pre-training data. Both Lauscher et al. [10] and Pires et al. [15] report cross-lingual transfer to be more successful when the source and target languages are typologically similar.

Linguistic typology is the study of which linguistic properties are exhibited by which languages [4]. Choenni & Shutova [2] investigate how typology is encoded in the embedding space of M-BERT, among others. They show that the model carves its embedding space into language-specific subspaces during pre-training. That is, sentence embeddings in a given language tend to cluster around a language-specific centroid. Moreover, probing experiments reveal that typological properties are encoded jointly across

languages. For example, subtracting the Modern Standard Arabic centroid from a Scottish Gaelic sentence embedding would remove the information from the embedding that verbs precede their subject. This indicates overlapping subspaces for typologically similar languages, which could be what underlies the relation between typological similarity and successful cross-lingual transfer.

The typologically-informed overlap between language-specific subspaces in a language model’s embedding space could be what underlies cross-lingual transfer, especially since cross-lingual transfer seems to be more successful for typologically similar languages. The goal of this project is to examine that intuition. This is done in a three-stage setup where the model is consecutively finetuned to a high-resource source language, one of various auxiliary languages, and finally for a few examples to one of various low-resource target languages. The specific aim is then to answer the following questions:

- Q1:** Does greater typological similarity between the source and target languages result in better zero-shot cross-lingual transfer performance?
- Q2:** To what degree does few-shot learning improve performance over zero-shot transfer?
- Q3:** Is few-shot transfer from a source to a target language more effective when first fine-tuning on a more typologically similar auxiliary language?
- Q4:** Does fine-tuning on a language make its typological features more easily identifiable in the model’s embedding space?

2 Related Work

Multilingual BERT (M-BERT) [5] is a transformer-based language model that has been fine-tuned on monolingual Wikipedia corpora from 104 languages, learning to map text from each of those languages to a shared embedding space. Pires et al. [15] investigated its cross-lingual generalizability by doing zero-shot transfer. Quantifying typological similarity as the number of shared word-order related features, they find typological similarity to be positively correlated with zero-shot performance. Lauscher et al. [10] similarly find zero-shot transfer to be most effective across typologically similar languages, but find that few-shot transfer – training on as few as 10 examples in the target language – yields considerable improvement over zero-shot transfer.

Further improvements in a low-resource setting can be obtained by adding an intermediary step with auxiliary languages. Nooralahzadeh et al. [14] use meta-learning as an intermediary step for improved performance on the tasks of natural language inference (NLI) and question answering (QA). In their setup, they first finetune M-BERT on a large, English-language dataset for the task at hand. Next, they use meta-learning to optimize the model parameters for quick adaptation to new languages. In short, it involves taking model parameters θ from the first stage, fine-tuning on an auxiliary language i to obtain θ_i , and taking the derivative of the loss $\mathcal{L}(\theta_i)$ with respect to the original parameters θ to update θ with. The result is a set of parameters optimized for adaptation to a new language. While they find this setup leads to improved zero-shot performance overall, they note that languages which share certain typological features benefit from this setup most.

Langedijk et al. [9] adapt this setup by using a typologically diverse set of auxiliary languages and focus on the task of dependency parsing. They find improvements over a zero-shot baseline are largest for target languages that are typologically distant from the source language, possibly because the auxiliary languages provide the model with new linguistic information that can later be leveraged. Choenni et al. [1] add language-specific subnetworks to the meta-learning setup. That is, they identify which subsets of parameters are most used when dealing with a given language, and restrict the model to only use that subset when processing that language. Similarly, they find that this setup yields greater improvements for typologically dissimilar source-target pairs. However, a setup that uses subnetworks without using meta-learning found the opposite effect: better improvements for more similar source and target languages.

Merchant et al. [12] investigate how the information encoded by BERT shifts during finetuning. In particular, they train probes to identify various linguistic features from a learned scalar mixing of the representations from each of BERT’s layers. They then compare the performance of these probes on a pre-trained BERT with the performance on a BERT-based model finetuned for NLI, QA, and dependency parsing. While finetuning typically does not alter the presence of these features BERT’s embeddings much, finetuning for dependency parsing causes syntax-related features to be more clearly identifiable.

Meanwhile, as stated, Choenni & Shutova [2] show that M-BERT carves its embedding space into language-specific subspaces, with typological features encoded jointly across languages. Combining this with Merchant et al.’s [12] findings, it is conceivable that finetuning on a specific language amplifies typological features relevant to that language in the model’s parameters. In turn, these features then become more readily available to any target language that shares them, which could explain the effectiveness of multistage cross-lingual transfer.

3 Methodology

The methodology involves a three-stage finetuning setup outlined in Section 3.2, with the model being fine-tuned for dependency parsing in a variety of languages. The choice for dependency parsing is based on the availability of uniform datasets in a wide variety of languages through Universal Dependencies (UD) [13]. The selection of languages is clarified in Section 3.3, the specifics of the model are discussed in Section 3.1.

3.1 Model

The model used is derived from UDify, [8] which is itself based on M-BERT [5]. UDify incorporates a layer attention mechanism that scales each layer for its importance to the task at hand. That is, during fine-tuning, a weighted sum $\mathbf{r}_j$ is computed for input token j over all layers $i \in [1, \dots, 12]$ as follows:

$$\mathbf{r}_j = \eta \sum_i \mathbf{U}_{i,j} \cdot \text{softmax}(\lambda)_i$$

where $\mathbf{U}_{i,j}$ is the output of layer i at token position j , η a trainable scalar, and λ a vector of trainable scalar mixing weights. Next, $\mathbf{r}_j$ is passed through two separate biaffine attention classifiers [6] for predicting the arc and the tag, respectively. Finally, the Chu-Liu/Egmonds algorithm [3] is used to obtain a valid dependency tree.

3.2 Fine-tuning setup

As stated, the fine-tuning setup follows three stages:

- Stage 1: The base model M_{base} is finetuned on the a treebank in the source language, German, to instill general knowledge of the task at hand, the result is M_{deu} ;
- Stage 2: M_{deu} is finetuned separately on treebanks of each of the auxiliary languages $\ell \in \{nld, nor, ces, fin\}$ ¹ to obtain four instances of M_ℓ ;
- Stage 3: M_{deu} and each M_ℓ is finetuned on each of the target languages $\lambda \in \{gsw, fao, hsb, vep\}$,² resulting in 4 instances of $M_{deu,\lambda}$ and 16 instances of $M_{\ell,\lambda}$.

So far, preliminary experiments have shown most updates to occur primarily in the classifier head and not in the M-BERT base. Given that the focus of this thesis is the cross-lingual generalizability of M-BERT, if the answer to **Q3** is that M-BERT’s embedding space remains largely unchanged throughout finetuning, that would more likely be the result of the particular setup used than of any property inherent to M-BERT. Therefore, two setups are ran alongside each other. In one, the classifier is updated in all finetuning steps. In the other, the classifier head is frozen after random initialization. The latter setup forces all updates to M-BERT’s parameters as its final embedding has to encode all information needed for a fully language-neutral, untrained classifier to parse dependencies in a given language. Should either setup result in findings that do not answer the research questions, it will be dropped in the thesis.

3.3 Languages

Broadly speaking, language selection is based on their typological similarity and their prevalence in M-BERT’s pre-training data as reported by Wu & Dredze [17]. Typological similarity is defined using the syntactic feature-vectors (`syntax_knn`) from the URIEL knowledge base [11]. In particular, each language is represented as a feature-vector, and similarity between languages is taken to be the cosine similarity between their respective vectors.

German was selected as one of the most prominent languages in M-BERT’s pre-training data. Moreover, Turc et al. [16] found German (alongside Russian) to be the most effective language for cross-lingual transfer. The selection of the auxiliary languages is motivated phylogenetically, as they each represent a different level of relatedness to German (Finnish is unrelated, Czech is in the same family, Norwegian is in the same branch, Dutch is in the same subbranch). However, a hierarchical clustering analysis of the `syntax_knn` vectors (Appendix A) found their selection to be typologically valid, too. The target languages were selected as being the most typologically similar language

¹ Dutch, Norwegian, Czech, and Finnish.

² Swiss German, Faroese, Upper Sorbian, and Veps.

to each of the auxiliary languages that (a) have a UD treebank and (b) are not part of M-BERT’s pretraining data. For Dutch, the most similar low-resource language was found to be Swiss German, Norwegian was matched with Faroese, Czech with Upper Sorbian, and Finnish with Veps.

For hyperparameter tuning, out of all languages with a treebank with more than 10 000 examples, the one whose URIEL vector is most equidistant to each of the other language vectors is used. This was found to be Portuguese. An overview of the languages, as well as the specific treebanks used is given in Appendix B.

4 Experimental Setup

The research questions can be classified as evaluating the success of this setup or, more broadly, of typologically guided cross-lingual transfer (**Q1**, **Q2**, **Q3**) and as investigating how shifts in the model parameters facilitate cross-lingual transfer (**Q4**). Section 4.1 outlines how the former group is answered, Section 4.2 does the same for the latter. Table B in Appendix B lists the specific datasets used in the experiments.

4.1 Model Evaluation

Successful transfer is defined as higher performance on the target language test set. Following the example of Lauscher et al. [10], the unlabeled attachment score (UAS) is used to quantify performance.³ In this section, $\text{UAS}(M_{base}, L)$ is taken to be the UAS of M_{base} evaluated on the test set of language L . Cosine similarity between URIEL’s `syntax_knn` vectors for languages $L1$ and $L2$ is denoted as $s(L1, L2)$. Moreover, as stated previously, ℓ is used for any of the auxiliary languages, λ is used for any of the target languages.

Q1 Does greater typological similarity between the source and target languages result in better zero-shot cross-lingual transfer performance?

If typological similarity indeed leads to better zero-shot performance, the expectation is that $\text{UAS}(M_{\ell}, l) - \text{UAS}(M_{base}, l)$ is higher when $s(\ell, l)$ is higher. That is, finetuning on German improves zero-shot performance more for typologically similar languages. Comparison with M_{base} is key, since models might generally struggle with certain treebanks regardless of typology. Here, l is any language in the union of auxiliary and target languages.

Q2 To what degree does few-shot learning improve performance over zero-shot transfer?

While Lauscher et al. [10] found few-shot learning to yield considerable improvements over zero-shot performance, it is important to quantify the degree to which that holds in this setup. To eliminate presence in M-BERT’s pre-training data as a factor, this question is only answered using few-shot learning on the target languages λ .

³ Using the more widely reported LAS would inhibit comparisons between datasets, as different treebanks may have been labeled with different annotation schemes.

Following Stage 3, there are twenty models $M_{l,\lambda}$ with l now being either German or any of the auxiliary languages. For each, improvement induced by few-shot learning is defined as $\text{UAS}(M_{l,\lambda}, \lambda) - \text{UAS}(M_l, \lambda)$. This difference is expected to be positive. Differences between pairs of l and λ fall under the scope of **Q3**.

Q3 Is few-shot transfer from a source to a target language more effective when first fine-tuning on a more typologically similar auxiliary language?

This question can effectively be split into two components. Both are answered by investigating whether $\text{UAS}(M_{l_1,\lambda}, \lambda) > \text{UAS}(M_{l_2,\lambda}, \lambda)$ whenever $s(l_1, \lambda) > s(l_2, \lambda)$. That is, few-shot performance is higher when the language finetuned on before few-shot learning on λ is more typologically similar. One component of the question pertains to the effectiveness of the auxiliary stage in general. In this case, l_1 is any auxiliary language ℓ and l_2 is German. When ℓ is more similar to λ than German is, the auxiliary stage may provide the model with new typological information that can then be leveraged during few-shot learning. The other component of the question pertains to the typological distance between the auxiliary language and the target language. In that case l_1 is any given auxiliary language and l_2 is any of the other auxiliary languages. This serves to assess whether any improvements found are the result of typology, rather than longer training regardless of typology.

4.2 Probing Experiments

Following Choenni & Shutova [2], investigation of the models' parameters is done using probing. In particular, they probe model embeddings for the presence of typological features. They first extract 10 000 random sentences from the Tatoeba corpora for a number of different languages. Each sentence is passed through the language models they investigate, they use mean-pooling over the token embeddings to obtain a sentence embedding. The embedding is then passed through a one-hidden-layer MLP with 100 units, followed by an n -unit softmax output layer, where n is the number of possible options for a given typological feature.

For this thesis, a similar approach is used. However, since Tatoeba only has 404 sentences in Veps, 404 sentences are sampled for each of the nine languages used to keep the dataset balanced. The exact probing task is described below.

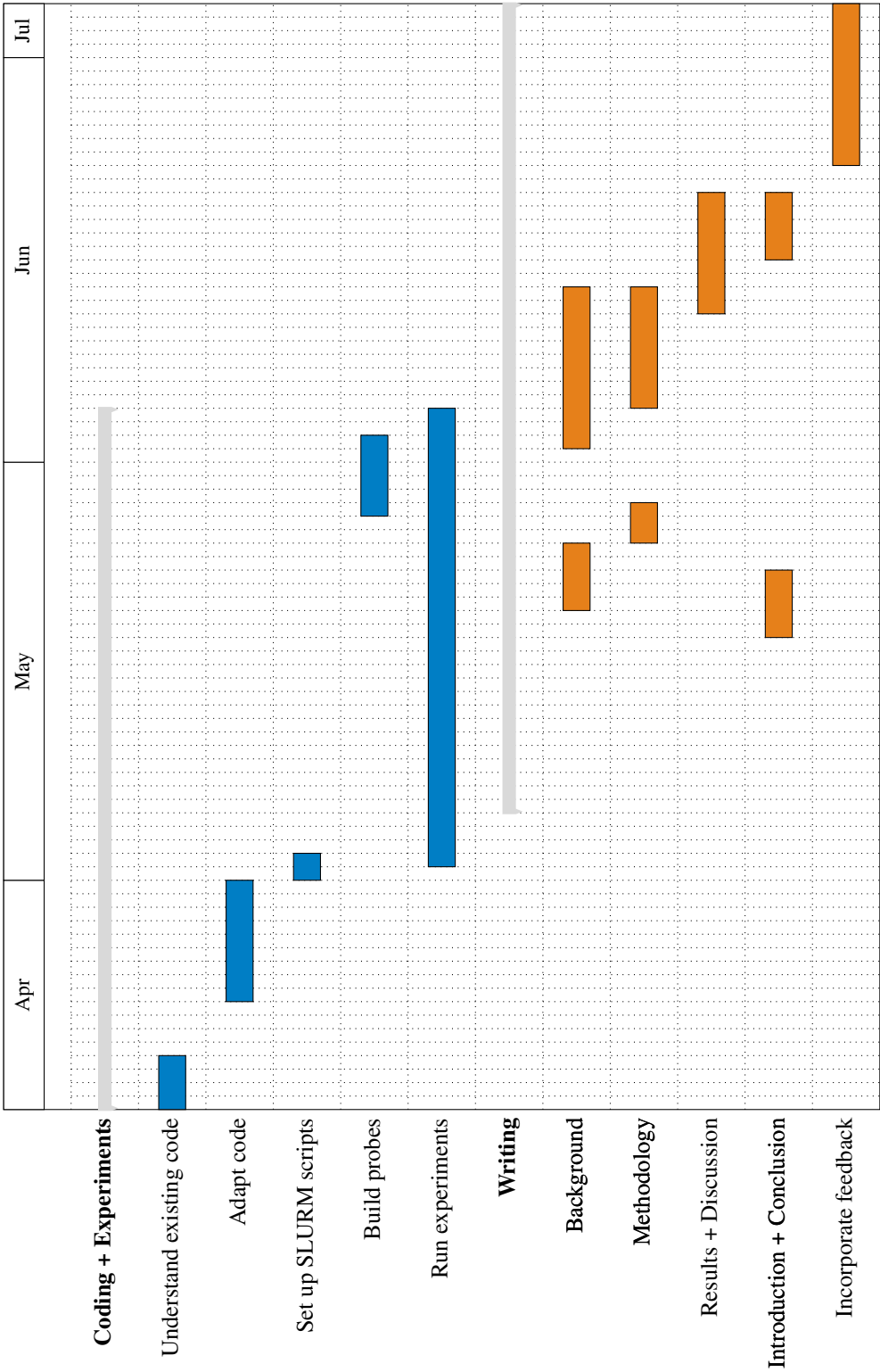
Q4 Does fine-tuning on a language make its typological features more easily identifiable in the model's embedding space? For this question, identifiability is defined by probe accuracy on the detection of typological features of a given language. The input to the probe is the sentence embedding obtained by mean-pooling over the token embeddings of the final layer of the M-BERT model. What constitutes the typological features to be identified may still be subject to change.

Initially, the probe is trained for language classification. That is, the probing task is a 9-class classification problem where the probe is given a sentence from any of the languages used and it is tasked with identifying the correct language. In this case, each language is viewed as a set of its typological features, and how easily identifiable its features are is quantified by how easily identifiable the language as a whole is. **Q4** would

then be answered in the affirmative when the probe attains a higher accuracy on language l compared to the other languages when applied to M_l .

If time permits, a separate probe might serve to identify specific typological features. The hidden layer would then be followed by separate binary classifiers for each feature in the URIEL `syntax_knn` feature vectors. This would more closely mirror the approach taken by Choenni & Shutova [2]. It may well be the case that cross-lingual transfer is not found to be effective between a pair of very similar languages. Should the feature-level probe find that the features shared by that pair are not encoded by the model, that would help explain why transfer is inhibited between the two. That said, analysis of these findings would be more time-consuming than those of the language-identification probe, additional caveats of using `syntax_knn` feature vectors notwithstanding. Therefore, feature-level probing will only be attempted after language-level probing has been implemented and sufficient time remains.

5 Planning



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A Appendix

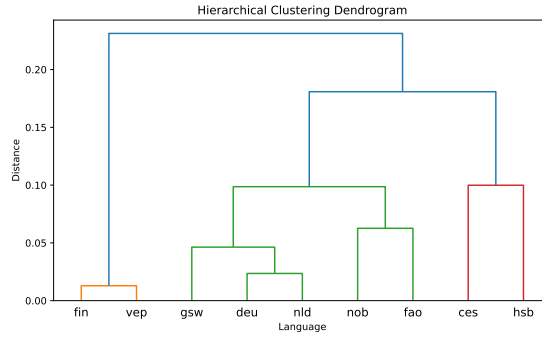


Fig. 1. Agglomerative Hierarchical Clustering dendrogram based on the cosine similarity of the `syntax_knn` vectors of the included languages.

B Appendix

Language	Family	Treebank	Number of tokens	Number of sentences
Czech	West Slavic	CAC	494 142	24 709
Dutch	West Germanic	Alpino	208 745	13 603
Faroese	North Germanic	OFT	10 002	1 208
Finnish	Finnic	TDT	202 210	15 136
German	West Germanic	HDT	3 399 390	189 928
Norwegian	North Germanic	Bokmaal	310 221	20 044
Portuguese	Romance	GSD	12 020	296 169
Swiss German	West Germanic	ATB	652	98
Upper Sorbian	West Slavic	UFAL	11 196	646
Veps	Finnic	VWT	1 303	103

Table 1. The specific UD treebanks featured in the project. Portuguese GSD is only used for hyperparameter tuning.